QUANTUM
SOURCE

Scaling Quantum Computing

Making scalable useful quantum computers a reality



Founding Team



Oded Melamed
CEO

Accomplished entrepreneur with more than 30 years of experience. Was CEO and co-founder of Altair Semiconductor, acquired by Sony in 2016 for $212 million.

BSc and MSc (Cum Laude) in Electrical Engineering from Tel Aviv University and an Executive MBA from the Kellogg-Recanati program.



Gil Semo
VP R&D

Experienced executive, with more than 25 years experience. Was Director VLSI and part of the founding team of Anobit (acquired by Apple in 2012).

After the acquisition was Director of System Architecture and a member of Apple Israel's management team. BSc in Electrical Engineering and MBA from Ben-Gurion University.



Prof. Barak Dayan
CHIEF SCIENTIST

Professor at the Weizmann Institute of Science, head of the Quantum Optics lab.

As a postdoc at Caltech, led the Chip-Based Cavity-QED lab in Professor H. J. Kimble's Quantum Optics Group.

Single-photon routing by a single atom
Science (2014)

Single-photon generation by a single atom on a fiber-coupled photonic chip Nature Photonics (2016)

First realization of a chip-based photon-atom quantum gate (SWAP)
Nature Physics (2018)



Dan Charash
CHAIRMAN

An accomplished entrepreneur with more than 30 years of experience. Was the CEO and co-founder of Provigent, acquired by Broadcom for over $300M.

MSc in EE from the Technion, and a BSc in Math and Physics from the Hebrew University and is a graduate of Israel's prestigious Talpiot program.

Building on 20 years of scientific leadership in photon-atom quantum gates

Caltech

Aoki, Dayan,.. & Kimble Nature (2006)
Dayan, Parkins,.. & Kimble Science (2008)
:

WEIZMANN

Shomroni,.. & Dayan Science (2014)
Rosenblum,.. & Dayan Nature Photonics (2016)
Bechler,.. & Dayan Nature Physics (2018)
:

QUANTUM SOURCE

QS Team PRX Quantum (2025)
QS Team arXiv:7574352 [quant-ph] (2026)
QS Team arXiv:2606.30385 [quant-ph] (2026)
:

QS by the numbers

Founded
2021

Seed and Round A
$77M

Patents
12

Employees
67

Hold PhD in Physics or EE
30

Professors
5

BACKED BY

pitango
GROVE VENTURES
DELL Technologies CAPITAL
ECLIPSE
standard investments
LEVEL
CANON EQUITY
10D

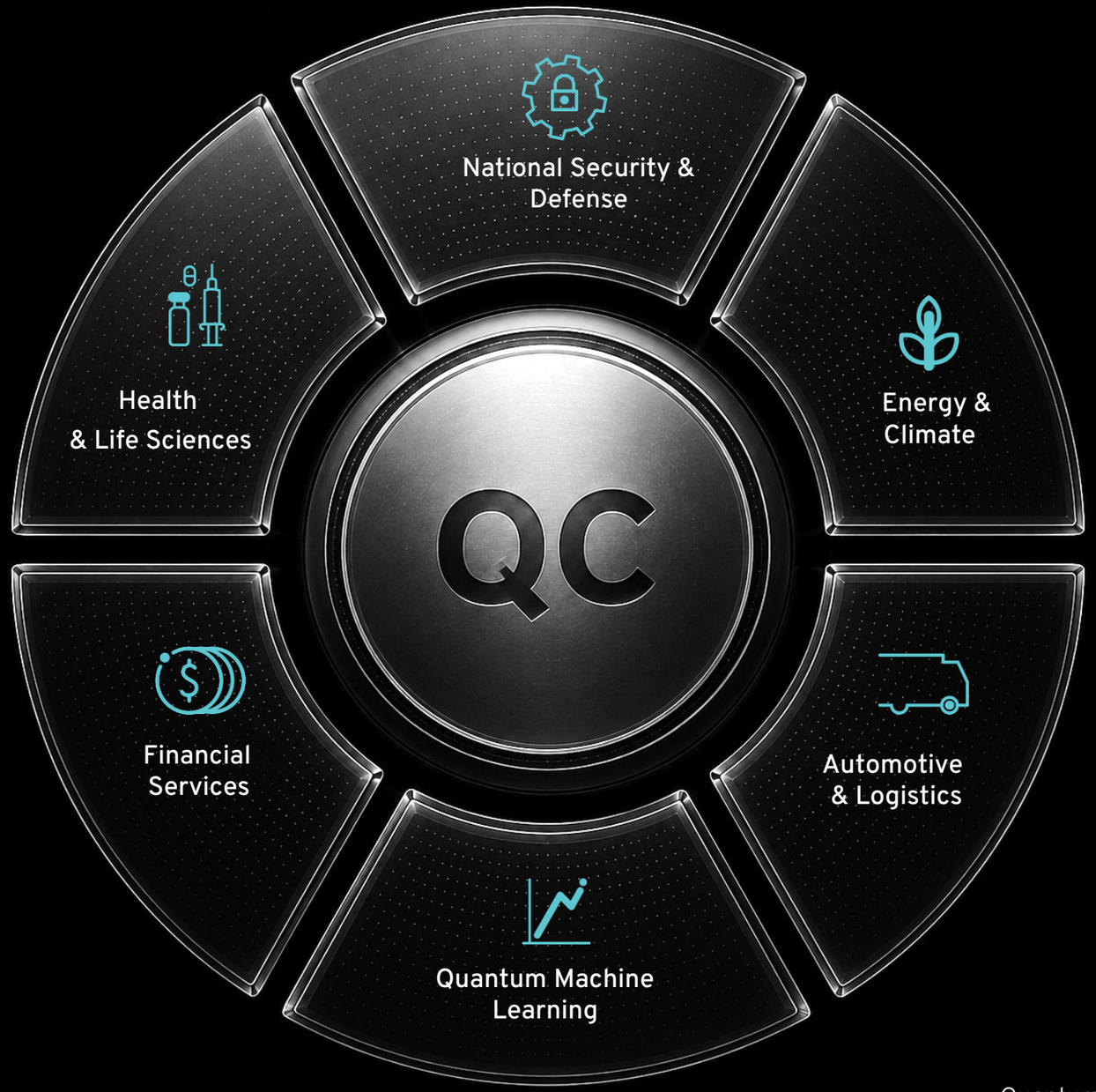
* The only Quantum investment of the world's largest classical computer company, Dell Technologies

The next
ECONOMIC FRONTIER

Quantum Computing will reach up to

$2 Trillion

World Changing Opportunity by 2035



Source: McKinsey & Co - Quantum Technology Monitor, Jun 2025

Large-scale, fault tolerant QC

needed to solve meaningful problems in a reasonable time



BREAKING RSA

1 million qubits
(~1500 logical qubits)



MOLECULE SIMULATION

0.5- 5 million qubits in
(700-7000 logical qubits)



DERIVATIVE PRICING

5+ million qubits
(8000+ logical qubits)

Error rate for logical qubit is better than 10^{-12}

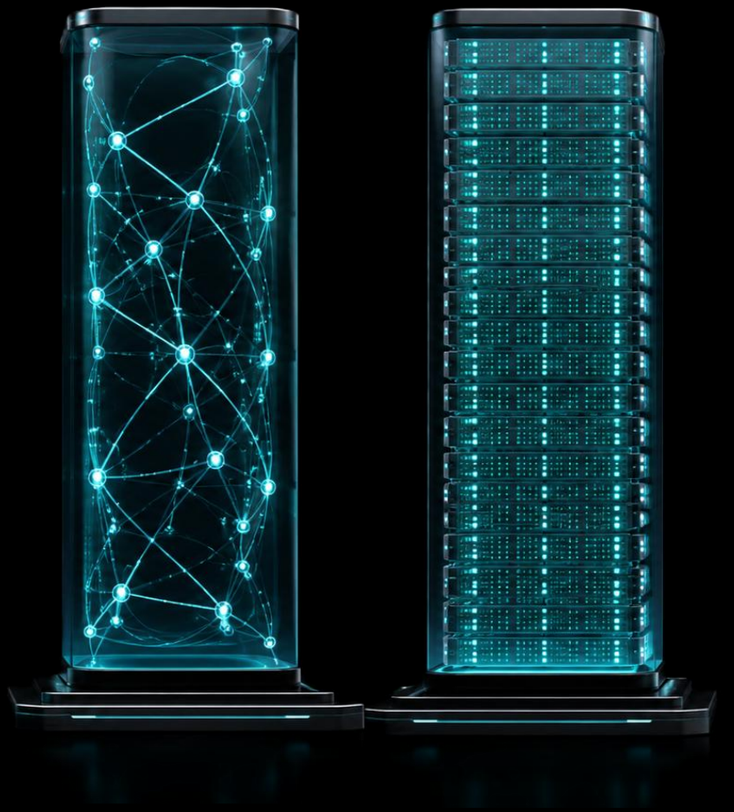
The Challenge

Two bars to clear

Millions of physical qubits running well below the error-correction threshold

&

With the connectivity to entangle qubits that are nowhere near each other



The Challenge

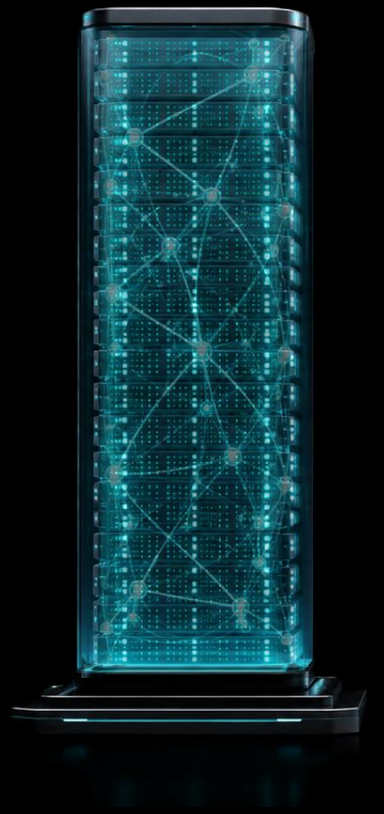
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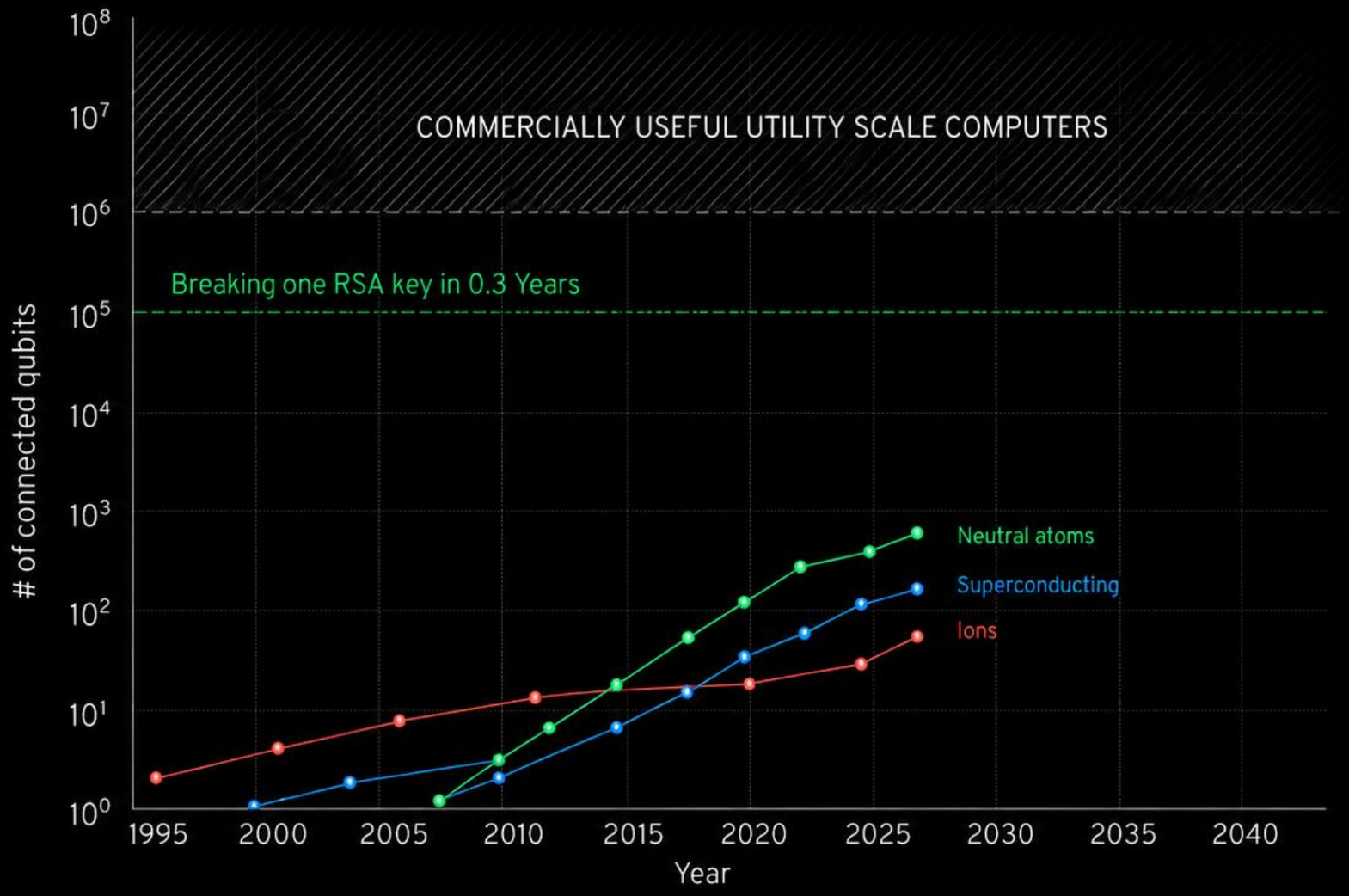
With the connectivity to entangle qubits that are nowhere near each other

SCALE = Numbers + Connectivity



Matter based qubits

Can't solve those two challenges: Scalability +Connectivity



Superconducting



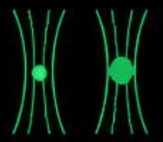
Planar chip layouts limit interactions largely to nearest neighbors; extending connectivity means added crosstalk and control complexity.

Trapped Ion



Shared motional modes crowd as more ions are added, degrading two-qubit gate fidelity as the register grows.

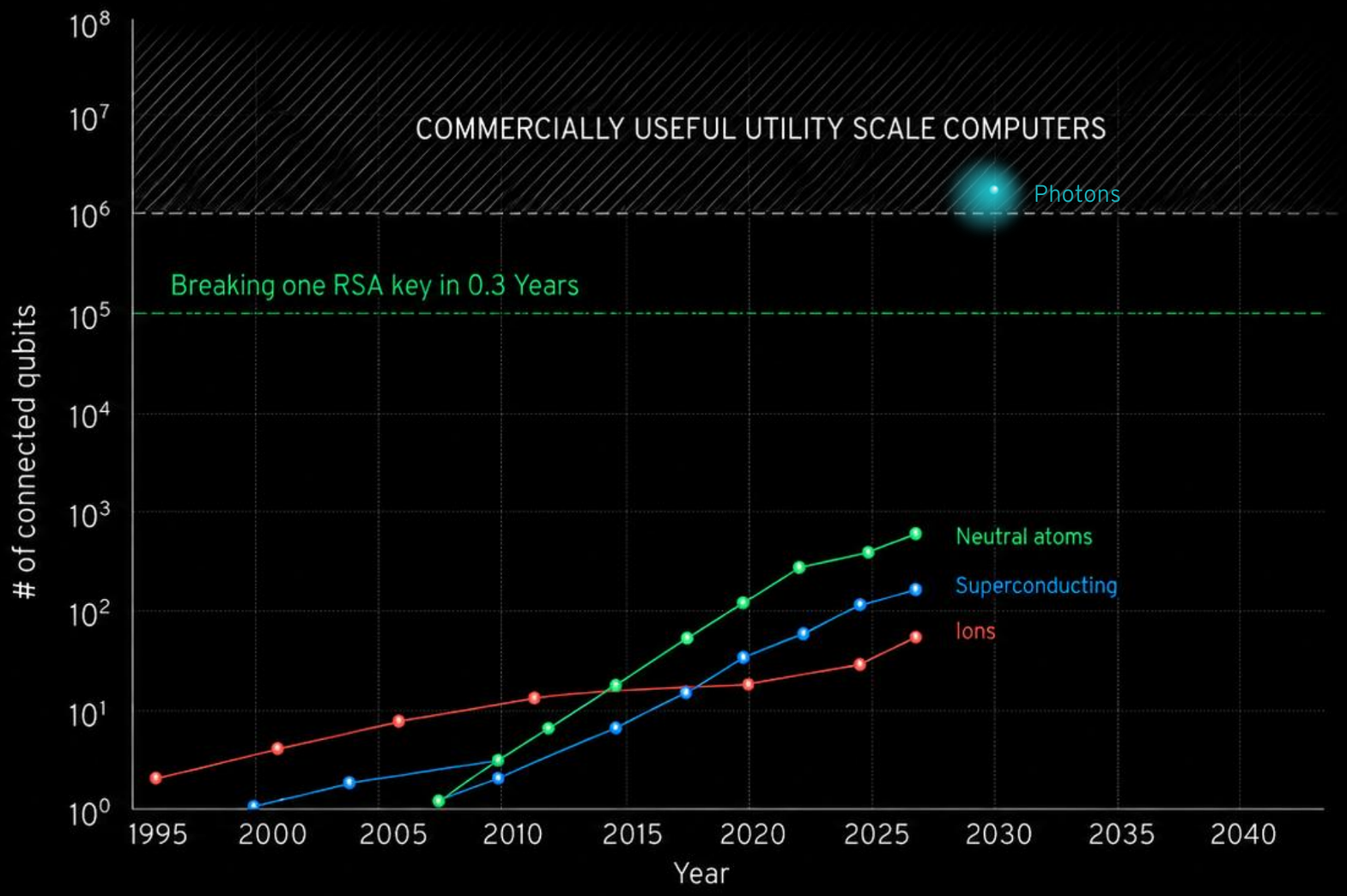
Neutral atom



Reconfigurable via shuttling, but mid-circuit moves and measurements push QEC rounds to milliseconds—1000× slower than superconducting.

Matter based qubits

Can't solve those two challenges: Scalability +Connectivity



Superconducting



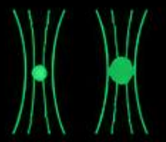
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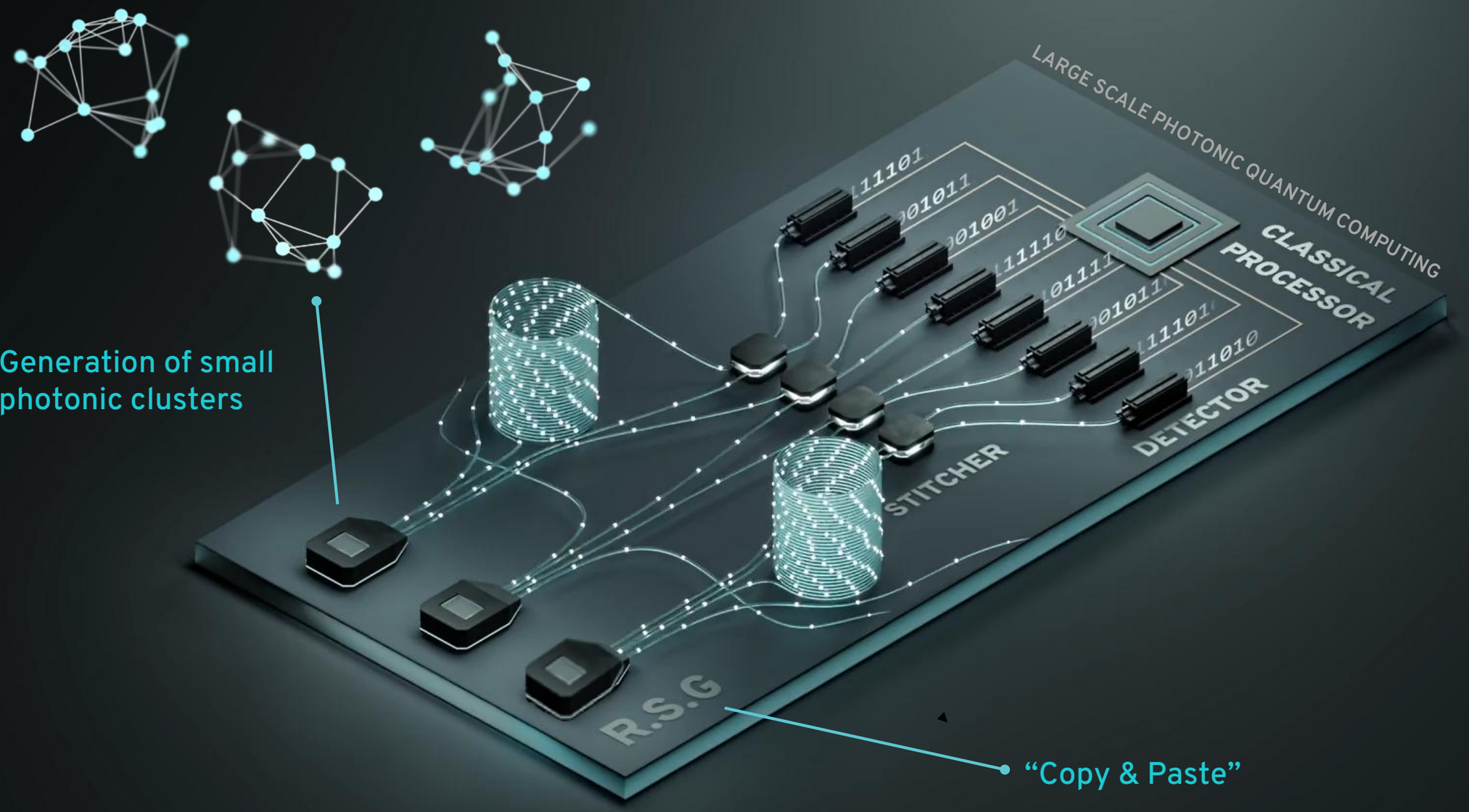
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Why Photons

Photons solve scale but create a new problem (SCALE = Numbers +Connectivity)

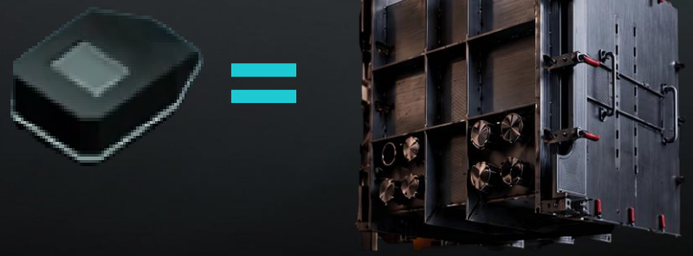


Why Photons

Photons solve scale but create a new problem



RSG



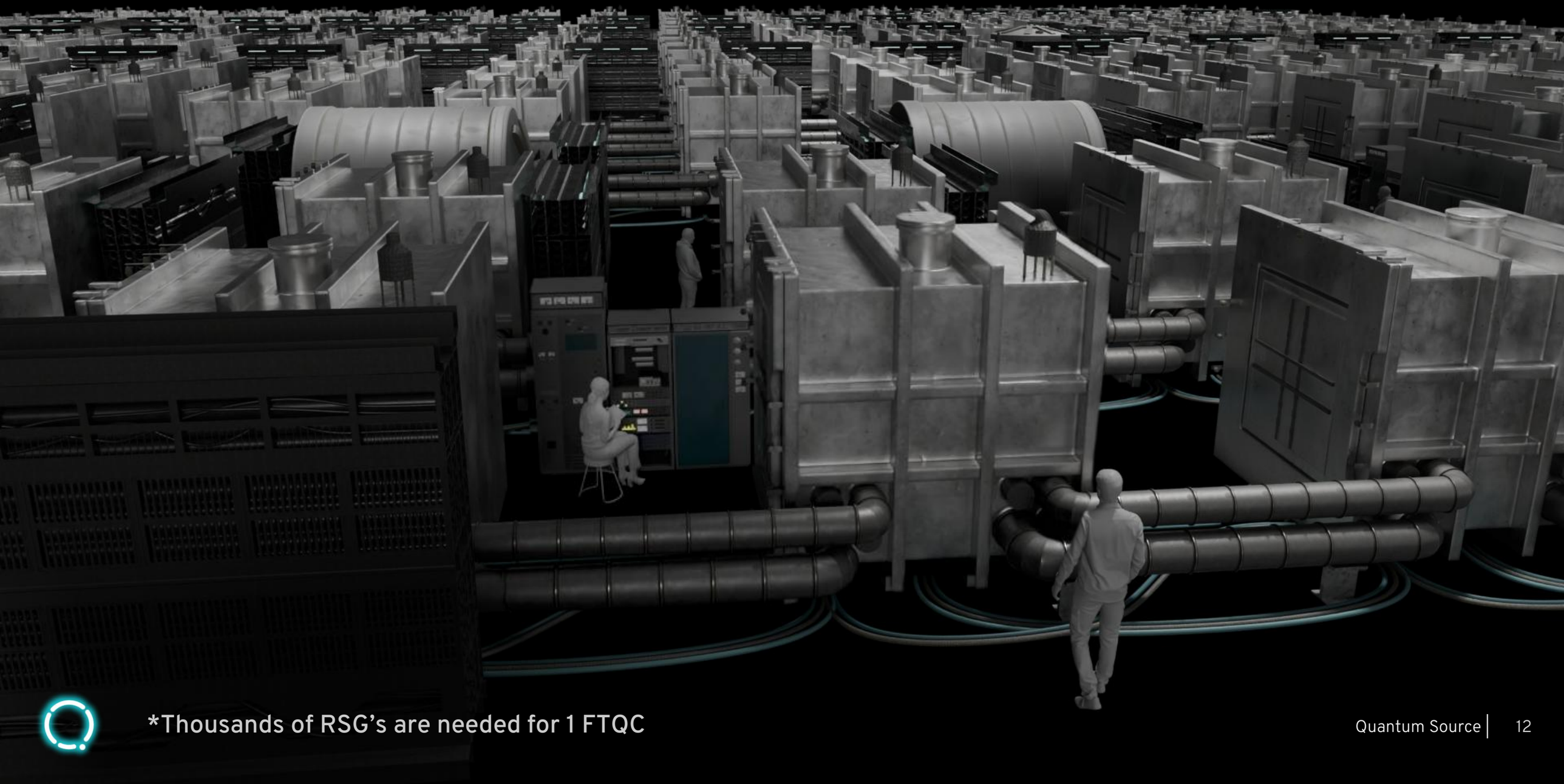
MASSIVE INEFFICIENCY

Using probabilistic photon generation & entanglement: **Efficiency 10^{-6}**

Thousands of RSG's are needed for 1 FTQC



Resource state generation



*Thousands of RSG's are needed for 1 FTQC

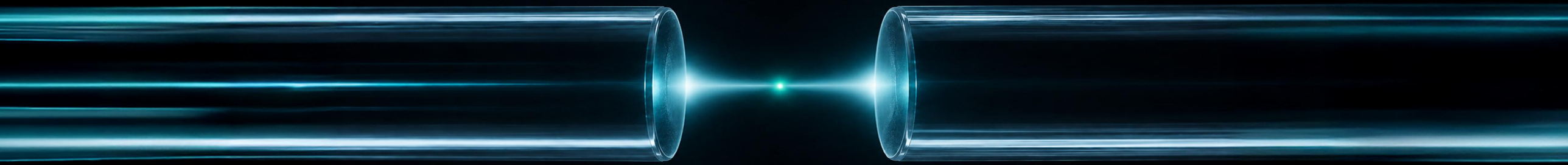
Quantum Source is solving this drastic inefficiency

By enabling deterministic interaction between single atoms and single photons

Inventing new modality >



THE COMPOUND-QUBIT MODALITY



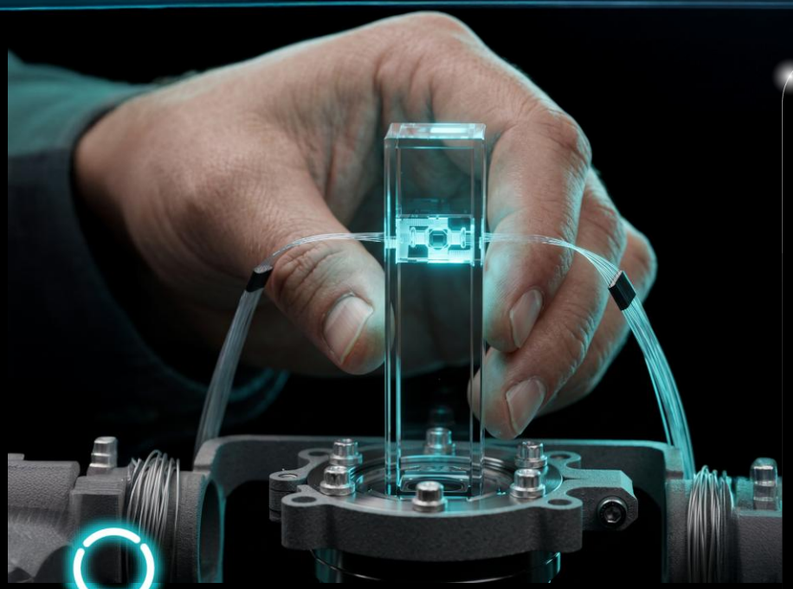
THE MODALITY
THAT RESOLVES THE SCALE CHALLENGE EFFICIENTLY

Our Advantage

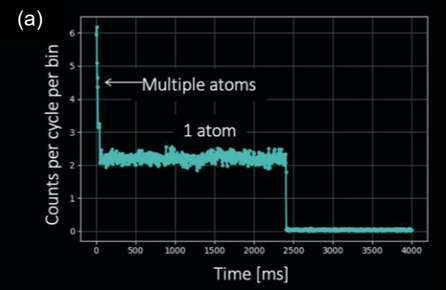
Deterministic Atom-Photon Gates

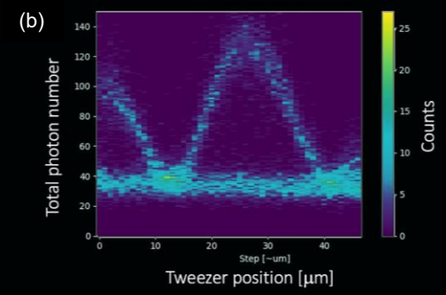
Confining light in micron-scale cavities creates strong coupling between single photons and single atoms

Single ^{87}Rb atom



- Seconds-long trapping with precise positioning of single atoms
- Optically coupled to the guided mode of the fiber





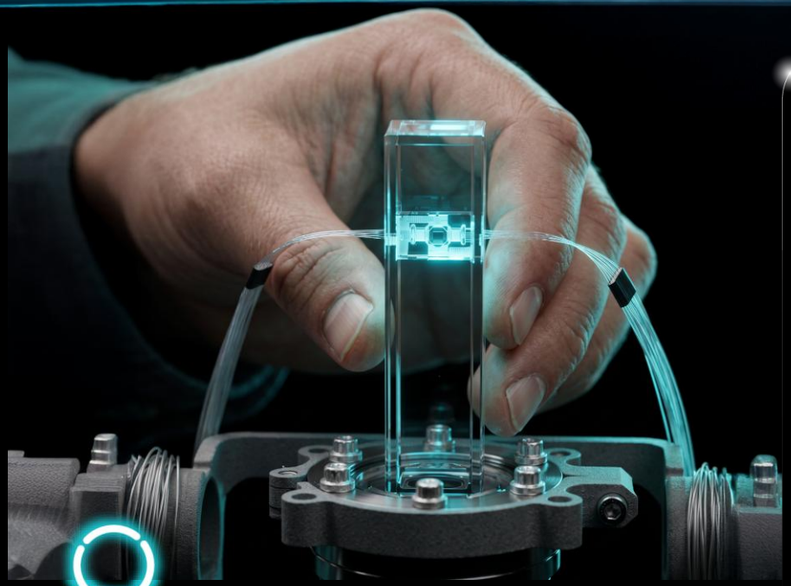
See also: arXiv:2605.09532v1 [quant-ph] – TO APPEAR IN NATURE PHOTONICS

Our Advantage

Deterministic Atom-Photon Gates

Confining light in micron-scale cavities creates strong coupling between single photons and single atoms





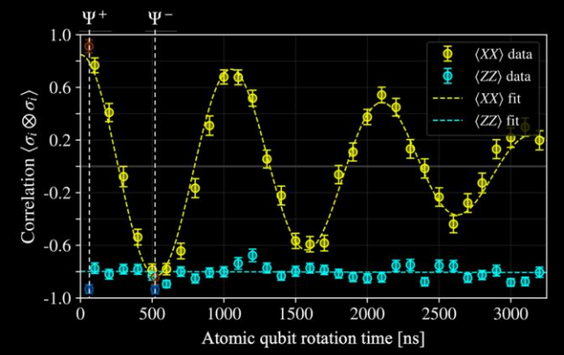
Enables DETERMINISTIC generation of entangled photonic states

Optically-controlled rapid atom-photon entanglement and Bell-state generation

Yoad Michael,[1,*] Omri Davidson,[1,*] Oded Zemer,[1] Yair Margalit,[1] Orel Bechler,[1] Dror Liran,[1] Noam Gross,[1] Doron Azoury,[1] Jeremy Raskop,[1] Nadav Kandel,[1] Geva Arwas,[1] Idit Peer,[1] and Barak Dayan[1,2,†]

[1] *Quantum Source Labs, Ilan Ramon 3 St., Ness Ziona, 7403636, Israel*

[2] *AMOS and Department of Chemical Physics, Weizmann Institute of Science, Rehovot, Israel.*



“Optically controlled rapid photon-atom entanglement and Bell-state generation”, to be submitted

Probabilistic VS Deterministic

Resource State Generator (RSG) is the key building block of photonic QC (95-99% of the BOM)
Thousands are needed for large-scale FT machine

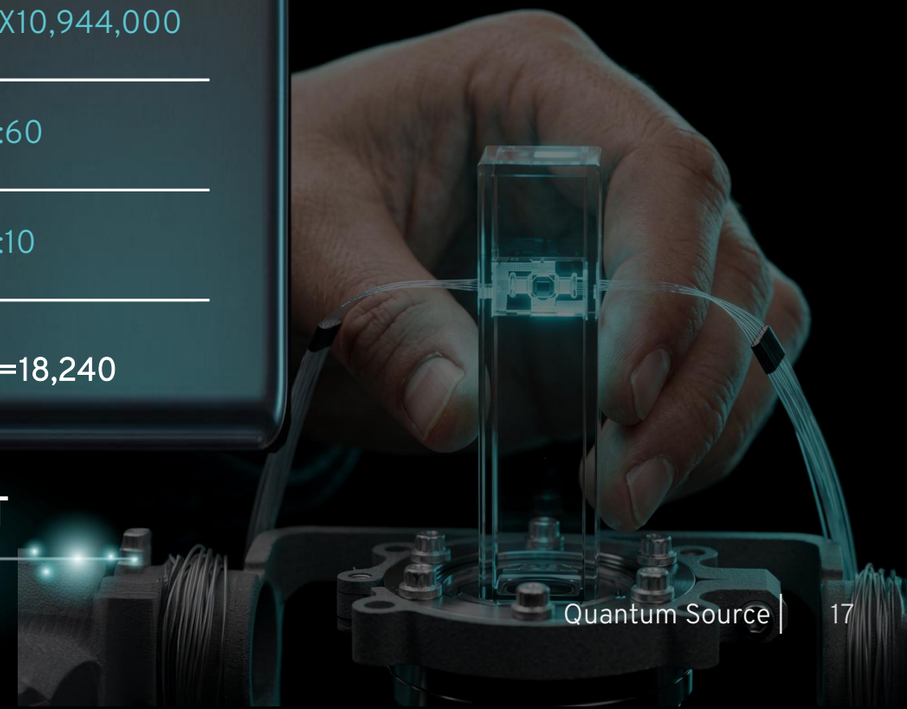


PROBABILISTIC

RSG | 18 - PHOTON RESOURCE STATE

DETERMINISTIC

2K	TEMPERATURE	Room Temp	
60,664,000	SOURCES	6	X10,944,000
1 ns	GENERATION TIME	60 ns	:60
1	FIBER LOSS FACTOR	10	:10
			=18,240



4 ORDERS OF MAGNITUDE MORE EFFICIENT

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4 ORDERS OF MAGNITUDE MORE EFFICIENT

Development stage

CORE PHYSICS IS PROVEN. NOW WE SCALE

PHYSICS FOUNDATION

- ✓ Atom trapping
- ✓ Photon-Atom entanglement
- ✓ Photon-Paire generation
- ✓ CZ Gate

CURRENT FOCUS

Production-ready physical layer

ENGINEERING SCALE-UP

MICRO DESIGN

UC miniaturization & high density detectors

FEATURES ENHANCEMENTS

Advanced routing & error correction protocols

ASSEMBLY

Utility Scale orchestration & control

FULL SYSTEM



Practical, Utility-scale Fault-tolerant Quantum computer

From physical layer to fault tolerance frame work

FTQC - High Level System Architecture

1 | On-demand photons are generated at the unit cells (UC)

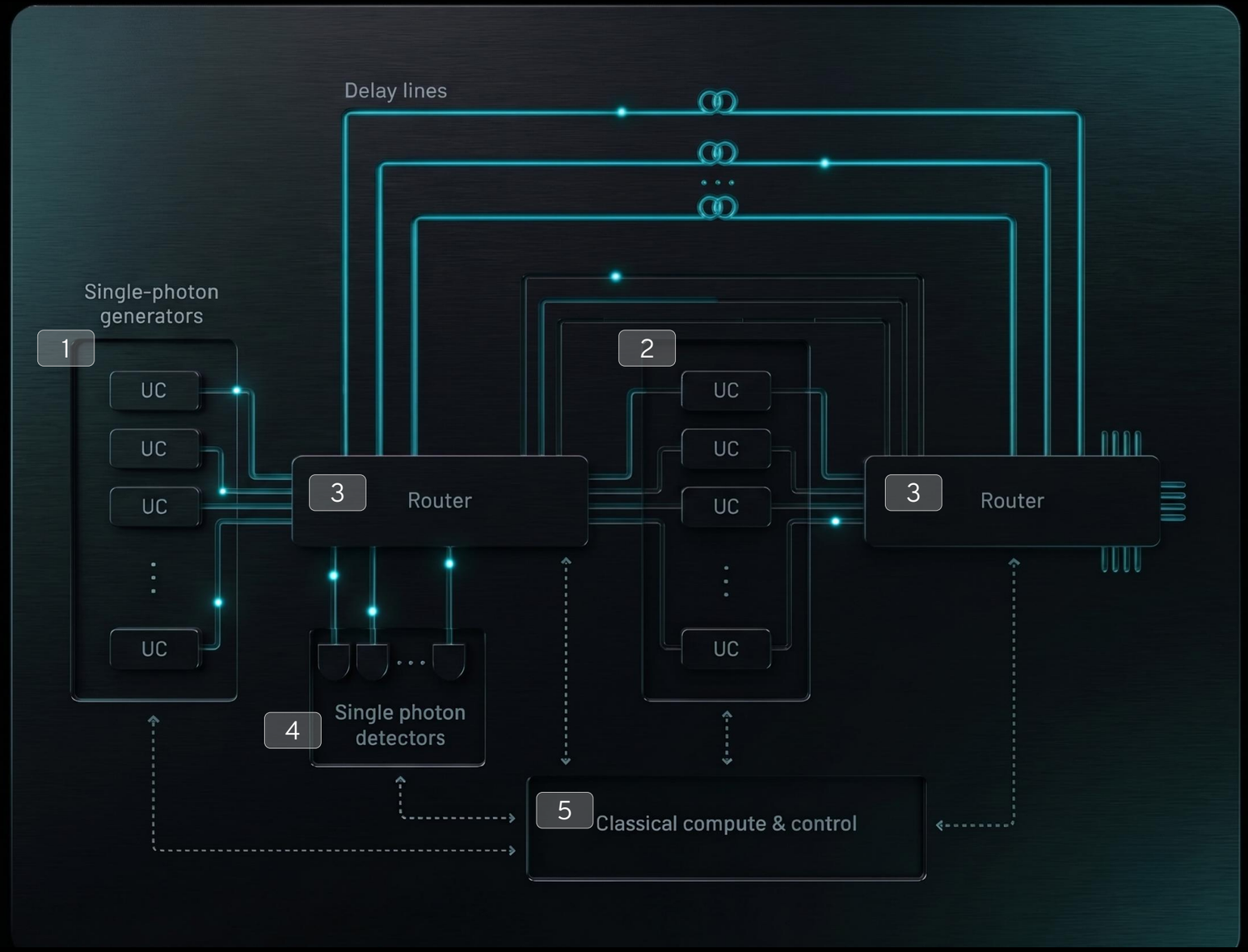
2 | They entangle with atoms at additional unit cells (UCs),

3 | Photons are routed to create cluster topologies

4 | Then detected by SNSPDs, building a measurement-based FTQC cluster state

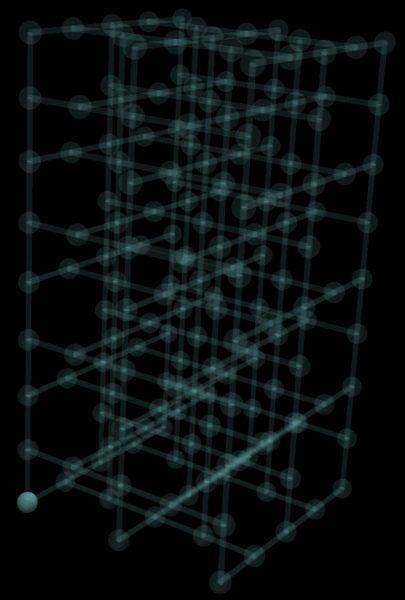
5 | Classical layer controls the system and decodes errors in real time

Scaling to any number of qubits, with simple duplication of the subsystems



QEC

Agile Quantum Error Correction



of qubits is not a hard limitation, we are optimizing for loss resilience

- Today this is RHG lattice, most resilient for photons loss

Highly agile architecture

- Modular routing allows adopting new codes with minimal modifications to the hardware.
- Advanced simulation environment allows testing and benchmarking new codes

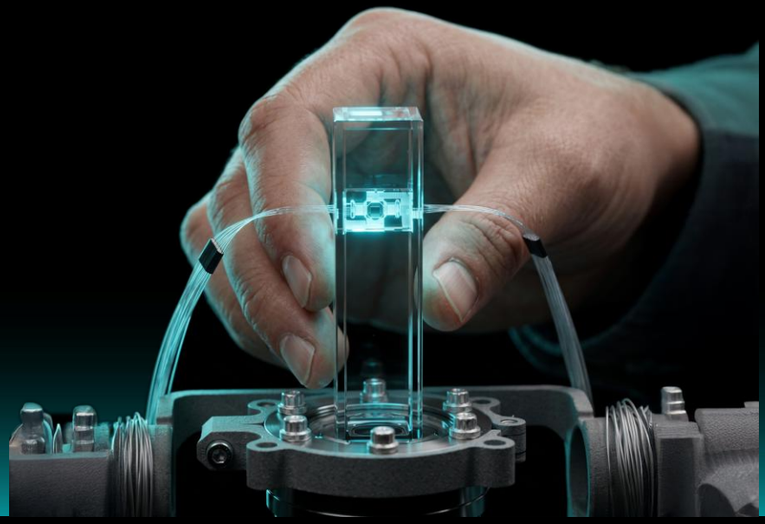
Fast error-correction cycles ● minimal routing overhead ● transversal gates

		Superconducting (Surface code)	Neutral Atoms (qLDPC)	Purely photonic (FBQC – topological code)	Quantum Source (RHG lattice)
Error-correction cycle		1 μSec	~1 $mSec$	1 μSec – 10+ μSec	200 $nSec$
Routing overhead		Lattice surgery & SWAP chains	Atom rearrangement & Logical teleportation	Lattice surgery & interleaving	Minimal overhead
Code selection		Limited by nearest-neighbor connectivity	Driven by # of qubits	Minimizing photon loss	Minimizing photon loss
RSA-2048 Benchmark	Runtime optimized	20Mq / 8 hours	100kq / 100 days	9700 RSG / 1 hour	8Mq / 1.5 hours
	HW footprint optimized	1Mq / 5 Days	10kq / 120 years	970 RSG / 9 hours	500kq / 30 hours

- Gidney and Ekera, "How to factor 2048-bit RSA integers in 8 hours using 20 million noisy qubits," Quantum 5, 433 (2021).
- Gidney, "How to factor 2048 bit RSA integers with less than a million noisy qubits," arXiv:2505.15917 (2025).
- Cain et al, "Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits," arXiv:2603.28627 (2026).
- Litinski and Nickerson. "Active volume: An architecture for efficient fault-tolerant quantum computers with limited non-local connections." arXiv:2211.15465 (2022).

Quantum Source Products

One core technology. Two product lines



ORIGIN

The Core Technology Building Block



QS-ORBIT

Large-scale, fault-tolerant quantum computer with a drastically reduced physical footprint and lower costs

QS-LINK

High-performance quantum interconnect, efficiently connects Quantum Processing Units (QPUs) from different modalities



QS-ORBIT Milestones

The roadmap to million-qubit systems

QS-ORBIT 0.5 — 2028

High-Quality Cluster

~50 physical qubits

Logical Error Rate: N/A

Loss per physical gate : <15%

QS-ORBIT 1 — 2029

Early FT System

~1K physical qubits

Logical Error Rate : 10^{-5}

Physical gate fidelity : 99.9%

Loss per physical gate : <7%

QS-ORBIT 2 — 2031

FTQC

100s to 1000s of logical qubits

~1K physical qubits per logical qubit

Logical Error Rate : 10^{-10}

Physical gate fidelity : 99.9%

Loss per physical gate : <5%

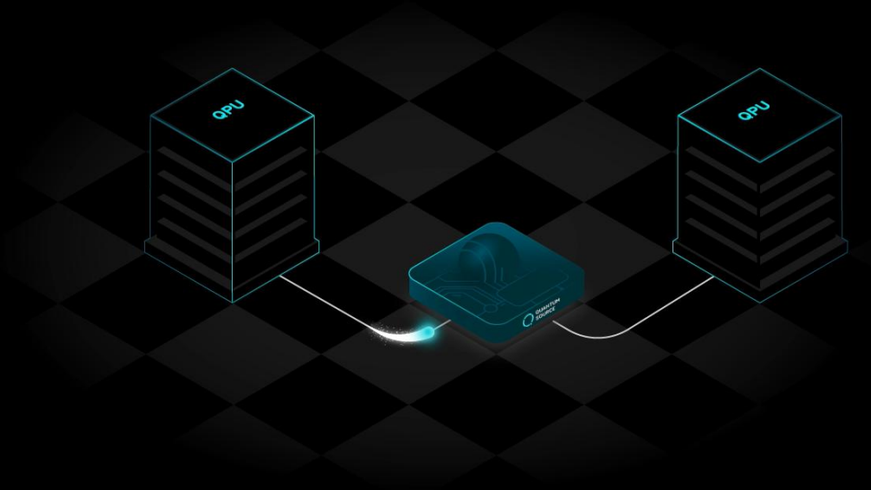
Error Correction & Hardware Scale

Mass Production



QS-LINK Solving Scaling Bottleneck of Matter Based QC

Highly Efficient Photonic Interconnection Solution



Matter-based scalability challenge calls for **Distributed Quantum Computing**, and **Photonic Interconnects** are emerging as the leading solution.

QS-LINK, based on a subset of our FTQC architecture, makes photonic interconnects deployable and effective.

*Availability by 2H/2027



- Data center scale connectivity, not limited to cm/m
- Heterogenous QPUs connectivity with the same QS-LINK hardware, no requirement for photons indistinguishability
- Room Temperature operation
- Practical entanglement rate, MHz compared to other solutions at KHz
- High entanglement success rate, 3 orders of magnitude better than alternatives

Manuscript in preparation

COMPETITIVE LANDSCAPE

	QUANTUM SOURCE	Photonic (i.e., PsiQuantum)	Superconducting (i.e., IBM, Google, Rigetti)	Trapped Ions (i.e., IonQ, Quantinuum)	Neutral Atoms (i.e., QuEra, Pasqal)
Operational temperature	**Room Temperature** 25°C	Cryogenic regime ~4 K (-269°C)	Cryogenic millikelvin regime ~10 mK (-273°C)	Room Temperature 25°C	Room Temperature 25°C
Scalability Potential	**>Millions**	Millions Large cluster-state photonic architectures	Thousands Scaling limited by cryogenic infrastructure	Thousands Limited by modular traps with ion shuttling	Tens of thousands Limited by optical infrastructure
Estimated System Footprint per logical qubit	**1 m²**	150 m² Cryogenic infrastructure and large photonic systems	50 m² Dilution refrigerator and cryogenic infrastructure	3 m² Laser control systems, optics and vacuum systems	5 m² Laser control systems, optical benches and vacuum chamber
RSA-2048 Benchmark: Runtime optimized	**8Mq / 1.5 hours**	9700 RSG / 1 hour	20Mq / 8 hours	Not published	100kq / 100 days
RSA-2048 Benchmark: Footprint optimized	**500kq / 30 hours**	970 RSG / 9 hours	1Mq / 5 Days	Not published	10kq / 120 years
Estimated Cost per physical qubit	**$40**	~$500-1,000 Large number of photonic circuits and cryogenic infrastructure	$10K - $100K Dilution refrigerator, Control electronics, Infrastructure and maintenance	$50K - $200K Ion trap hardware and lasers, optics & vacuum systems	~$5K - $20K High-power laser systems, precision optics and ultra-high-vacuum infrastructure

Logical qubit error rate is 10^{-12}

Best | Moderate | Weak

Summary

Scaling Quantum Computing



Full stack QC

We build the utility scale fault tolerant quantum computer

Massive Efficiency Advantage

Scaling up with an over x10,000 efficiency improvement compared to purely photonic approaches

Practical Cost For Enterprise Adoption

Making quantum computing economically viable, accelerating real-world deployment

Highly agile Error Correction architecture

Fast error correction cycles, with an architecture flexible enough to adopt new codes without redesigning the hardware

Differentiated Interconnect Solution

The same core technology powers highly differentiated interconnect solution that supports scaling and heterogeneous quantum computing architecture systems

QUANTUM SOURCE

Thank you

www.qs-labs.com

